1.9 More exercises

Example 14. *Define an appropriate language and formalize the following sentences using First Order Logic formulas.*

1. *Bill has at least one sister.*
2. *Bill has no sister.*
3. *Bill has at most one sister.*
4. *Bill has (exactly) one sister.*
5. *Bill has at least two sisters.*
6. *Every student takes at least one course.*
7. *Only one student failed Geometry.*
8. *No student failed Geometry but at least one student failed Analysis.*
9. *Every student who takes Analysis also takes Geometry.*

1. $\exists x.SisterOf(x, Bill)$
2. $\neg\exists x.SisterOf(x, Bill)$
3. $\forall x\forall y.(SisterOf(x, Bill) \wedge SisterOf(y, Bill) \rightarrow x = y)$
4. $\exists x.(SisterOf(x, Bill) \wedge \forall y.(SisterOf(y, Bill) \rightarrow x = y))$
5. $\exists x\exists y.(SisterOf(x, Bill) \wedge SisterOf(y, Bill) \wedge \neg(x = y))$
6. $\forall x.(Student(x) \rightarrow \exists y.(Course(y) \wedge Takes(x, y)))$
7. $\exists x.(Student(x) \wedge Failed(x, Geometry) \wedge \forall y.(Student(y) \wedge Failed(y, Geometry)$
8. $= y))$
9. $\neg\exists x.(Student(x) \wedge Failed(x, Geometry)) \wedge \exists x.(Student(x) \wedge Failed(x, Analysis))$
10. $\forall x.(Student(x) \wedge Takes(x, Analysis) \rightarrow Takes(x, Geometry))$

Example 15. *Consider the following sentences:*

1. *All actors and journalists invited to the party are late.*
2. *There is at least a person who is on time.*
3. *There is at least an invited person who is neither a journalist nor an actor.*

Formalize the sentences and prove that 3. is not a logical consequence of 1 . and 2.

Solution.

1. $\forall x \cdot ((a(x) \vee j(x)) \wedge i(x) \rightarrow l(x))$
2. $\exists x.\neg l(x)$
3. $\exists x.(i(x) \wedge \neg a(x) \wedge \neg j(x))$

It's sufficient to find an interpretation $\mathcal{I}$ for which the logical consequence does not hold:

	$l(x)$	$a(x)$	$j(x)$	$i(x)$
Bob	F	T	F	F
Tom	T	T	F	T
Mary	T	F	T	T

Example 16. *Let $\{c_1, .., c_k\}$ be a non empty and finite set of colors. A partially colored directed graph is a structure $\langle N, R, C\rangle$ where*

- *N is a non empty set of nodes*
- *R is a binary relation on N*
- *C associates colors to nodes (not all the nodes are necessarily colored,. and each node has at most one color)*

Provide a first order language and a set of axioms that formalize partially colored graphs. Show that every model of this theory correspond to a partially colored graph, and vice-versa. For each of the following properties, write a formula which is true in all and only the graphs that satisfies the property:

1. *connected nodes don't have the same color*
2. *the graph contains only 2 yellow nodes*
3. *starting from a red node one can reach in at most 4 steps a green node*
4. *for each color there is at least a node with this color*
5. *the graph is composed of $|C|$ disjoint non empty subgraphs, one for each color*

Solution. Language

- a binary predicate edge, where edge(n, m) means that node n is connected to node m
- a binary predicate color, where color (n, x) means that node n has color x

- the following constants: yellow, green, red

Axioms

0. "each node has at most one color"

$$\forall n \forall x.(\text{color}(n,x) \rightarrow \neg \exists y.(y \neq x \wedge \text{color}(n,y)))$$

1. "connected nodes do not have the same color"

$$\forall n \forall m \forall x.(\text{edge}(n,m) \wedge \text{color}(n,x) \rightarrow \neg\, \text{color}(m,x))$$

2. "the graph contains only two yellow nodes"

$$\begin{aligned}\exists n \exists n'.\,(\ \text{color}\ (n,\ \text{yellow}\ ) \wedge\ \text{color}\ \ (n',\ \text{yellow}\ ) \wedge n \neq n' \wedge \\ \forall m.\,(m \neq n \wedge m \neq n' \rightarrow \neg\, \text{color}(m,\ \text{yellow}\ )))\end{aligned}$$

3. "starting from a red node one can reach in at most 4 steps a green node"

$$\begin{aligned}&\forall n(\ \text{color}\ (n,\ \text{red}\ ) \rightarrow \\ &(\exists n_1 \cdot (\ \text{edge}\ \ (n,n_1) \wedge\ \text{color}\ \ (n_1,\ \text{green}\ )) \vee \\ &\exists n_1,n_2 \cdot (\ \text{edge}\ \ (n,n_1) \wedge\ \text{edge}\ \ (n_1,n_2) \wedge\ \text{color}\ \ (n_2,\ \text{green}\ )) \vee \\ &\exists n_1,n_2,n_3 \cdot (\ \text{edge}\ \ (n,n_1) \wedge\ \text{edge}\ \ (n_1,n_2) \wedge\ \text{edge}\ \ (n_2,n_3) \wedge \text{color}\,(n_3,\ \text{green}\ )) \vee \\ &\exists n_1,n_2,n_3,n_4 \cdot (\ \text{edge}\ \ (n,n_1) \wedge \text{edge}\,(n_1,n_2) \wedge \text{edge}\,(n_2,n_3) \wedge \text{edge}\,(n_3,n_4) \wedge \text{color}\,(n_4,\ \text{green}\ )) \\ &))\end{aligned}$$

4. "for each color there is at least a node with this color"

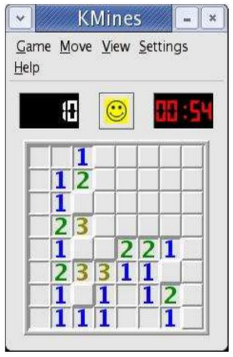
$$\forall x \exists n.\, \text{color}(n,x)$$

5. "the graph is composed of $|C|$ disjoint non empty subgraphs, one for each color"

$$\begin{aligned}&\forall x \exists n.\, \text{color}(n,x) \wedge \\ &\forall n \exists x.\, \text{color}(n,x) \wedge \\ &\forall n \forall x.(\text{color}(n,x) \rightarrow \neg \exists y.(y \neq x \wedge \text{color}(n,y))) \wedge \\ &\forall n \forall m \forall x.(n \neq m \wedge \text{color}(n,x) \wedge \text{color}(m,x) \rightarrow \\ &\left(\text{edge}(n,m) \bigvee_{i=1}^{|N|} \left(\exists n_1,..,n_i.\, \left(\text{edge}\,(n,n_1) \bigwedge_{j=1}^{i-1} \text{edge}\,(x_j,x_j+1) \wedge \text{edge}\,(n_i,m)\right)\right)\right)\end{aligned}$$

Example 17. *Minesweeper is a single-player computer game invented by Robert Donner in 1989. The object of the game is to clear a minefield without detonating a mine. The game screen consists of a rectangular field of squares. Each square can be cleared, or uncovered, by clicking on it. If a square that contains a mine is clicked, the game is over. If the square does not contain a mine, one of two things can happen: (1) A number between 1 and 8 appears indicating the amount of adjacent (including diagonally-adjacent) squares containing mines, or (2) no number appears; in which case there are no mines in the adjacent cells. An example of game situation is provided in the following figure:*
Provide a first order language that allows to formalize the knowledge of a player in a game state. In such a language you should be able to formalize the following knowledge:

1. *there are exactly n mines in the minefield*
2. *if a cell contains the number 1 , then there is exactly one mine in the adjacent cells.*



3. *show by means of deduction that there must be a mine in the position* $(3,3)$ *of the game state of picture 3.1*

Suggestion: define the predicate Adj(x, y) to formalize the fact that two cells x and y are adjacent

Solution. Language

1. A unary predicate mine, where mine (x) means that the cell xcontains a mine
2. A binary predicate adj, where adj(x, y) means that the cell x is adjacent to the cell y
3. A binary predicate contains, where contains(x, n) means that the cell x contains the number n

Axioms

1. There are exactly n mines in the game.

$$\exists x_1, ..\exists x_n \left(\bigwedge_{i=1}^{n} \text{mine}\,(x_i) \wedge \forall y \left(\text{mine}(y) \rightarrow \bigvee_{i=1}^{n} y = x_i \right) \right)$$

2. If a cell contains the number 1 , then there is exactly one mine in the adjacent cells.
$\forall x.(\text{contains}(x, 1) \rightarrow \exists z.(\text{adj}(x, z) \wedge \text{mine}(z) \wedge \forall y.(\text{adj}(x, y) \wedge \text{mine}(y) \rightarrow y = z)))$

3. Show by means of deduction that there must be a mine in the position $(3, 3)$
from Picture 3.1 we have:
a. contains((2,2),1)
b. $\neg\,\text{mine}((1, 1)) \wedge \neg\,\text{mine}((1, 2)) \wedge \neg\,\text{mine}((1, 3))$
c. $\neg\,\text{mine}((2, 1)) \wedge \neg\,\text{mine}((2, 2)) \wedge \neg\,\text{mine}((2, 3))$
d. $\neg\,\text{mine}((3, 1)) \wedge \neg\,\text{mine}((3, 2))$
we can deduce:
e. $\exists z \cdot (\text{adj}((2, 2), z) \wedge \text{mine}(z) \wedge \forall y \cdot (\text{adj}((2, 2), y) \wedge \text{mine}(y) \rightarrow y = z))$ $\rightsquigarrow$ from a. and axiom 2
f. mine $((1, 1)) \vee \text{mine}((1, 2)) \vee \text{mine}((1, 3)) \vee \text{mine}((2, 1)) \vee \text{mine}((2, 2)) \vee \text{mine}((2, 3)) \vee \text{mine}((3, 1)) \vee \text{mine}((3, 2)) \vee \text{mine}((3, 3))$ $\rightsquigarrow$ from e.
g. mine $((3, 3))$ $\rightsquigarrow$ from b.,c.,d. and f.

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Example 18. *Formalize in first order logic the train connections in Italy. Provide a language that allows to express the fact that a town is directly connected (no intermediate train stops) with another town, by a type of train (e.g., intercity, regional, interregional). Formalize the following facts by means of axioms:*

1. *There is no direct connection from Rome to Trento*
2. *There is an intercity from Rome to Trento that stops in Firenze, Bologna and Verona.*
3. *Regional trains connect towns in the same region*
4. *Intercity trains don't stops in small towns.*

Solution. *We define the language as follows*
Constants $RM, FI, BO, VR, TN, \ldots$ are identifiers of the towns of Roma, Firenze, Bologna, Verona, Trento, . . . and InterCity, Regional, . . . are the identifiers of the type of trains
Predicates Train with arity equal to 1 , where Train (x) means x is a train Town with arity equal to 1 , where Train (x) means x is a town
SmallTown with arity equal to 1, where Train (x) means x is a small town

TrainType with arity equal to 2 , where TrainType (x, y) *means that the train* x *is of type* y.
IsInRegion with arity equal to 2 , where IsInRegion (x, y *) means that the town* x *is in region* y. *DirectConn with arity equal to 3, where DirectConn* (x, y, z) *means that the train* x *directly connects (with no intermediated stops) the towns* y *and* z.

Background axioms With these set of axioms we have to formalize some background knowledge which is necessary to make the formalization more adeguate

1. *a train has exactly one train type;*

$$\forall x(\text{Train}(x) \rightarrow \exists y(\text{TrainType}(x, y))) \wedge \forall xyz(\text{TrainType}(x, y) \wedge \text{TrainType}(x, z) \rightarrow y = z) \tag{3.1}$$

2. *Intercity type is different from regional type:*

$$\neg(\ \textit{InterCity} = \textit{Regional}\ ) \quad (\ \textit{also written as InterCity} \neq \textit{Regional}\ ) \tag{3.2}$$

3. *A town is associated to exactly one region*

$$\forall x(\text{Town}(x) \rightarrow \exists y(\text{IsInRegion}(x, y))) \wedge \forall xyz(\text{IsInRegion}(x, y) \wedge \text{IsInRegion}(x, z) \rightarrow y = z) \tag{3.3}$$

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